

Benchmarking of NQCH's quantum computer

April 23, 2026

1. Report of Changes

Platform: sinq20

Calibration-id: 64983fcc115ea20314ac2d69244d40c2ce85b7e9

Calibration date: 2026-04-22 09:05:31

Calibration note: Merge pull request #16 from qiboteam/160426-coarsecal
160426 coarsecal

Experiment-id: 20260422212203

Experiment date: 2026-04-22 21:22:04

Experiment note: temporary note!!!

Platform: sinq20

Calibration-id: 4dc4082f38a53222b3956c22202d32a520d4bc78

Calibration date: 2026-01-15 02:03:03

Calibration note: chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

Experiment-id: 20260118161538

Experiment date: 2026-01-18 16:15:38

Experiment note: temporary note!!!

2. Version Comparison

Library	Version	Library	Version
qibo	0.2.23	numpy	2.4.4
qibolab	0.2.9	qibocal	0.2.5
matplotlib	3.10.8	scipy	1.17.1
scikit-learn	1.8.0	pandas	2.3.3
networkx	3.6.1	sympy	1.14.0
torch	2.11.0		

Library	Version	Library	Version
qibo	0.2.22	numpy	2.2.6
qibolab	0.2.9	qibocal	0.2.4
matplotlib	3.10.3	scipy	1.15.3
scikit-learn	1.6.1	pandas	2.2.3
networkx	3.4.2	sympy	1.14.0
torch	2.7.0		

3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



8. Mermin

Mermin's algorithm for 3 qubits.

Runtime	Qubits	Max
272.36 sec	1, 4, 5	-3.428



Runtime	Qubits	Max
1.02 sec	0, 2, 3	-3.316



9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in [[18, 14]].

Runtime	Qubits	Max prob
3.02 sec	18, 14	0.88



Runtime	Qubits	Max prob
2.36 sec	2, 3	0.89



10. Grover - 3 qubits

Grover's algorithm for 3 qubits executed on `siq20` backend with 1000 shots per circuit. We measure the success rate of finding the target state '111' for each pair of qubits in `[[1, 4], [4, 5]]`.

Runtime	Qubits	Max prob
3.22 sec	1, 5, 4	0.188



Runtime	Qubits	Max prob
2.43 sec	0, 2, 3	0.385



11. GHZ state preparation

GHZ circuit with 3 qubits executed on `siq20` backend with 1000 shots.

Runtime	Qubits	Success
3.00 sec	1, 4, 5	0.930



Runtime	Qubits	Success
3.95 sec	0, 3, 2	0.910

